



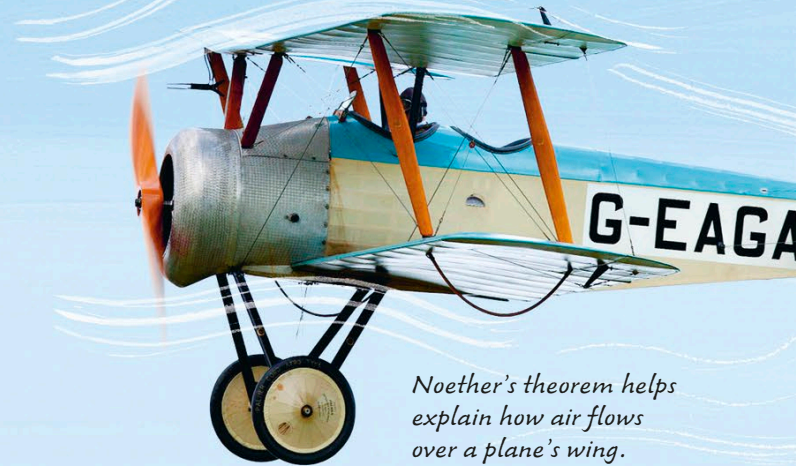
EMMY NOETHER

German mathematician
1882–1935

Noether's theorem explains why the laws of physics stay the same everywhere in the universe.

Emma Noether was determined to follow in her father's footsteps and become a mathematician. At that time in Germany, girls weren't allowed to study for a college degree. They could sit in on classes, though, which is exactly what Emmy did, at the University of Erlangen. She was one of only two women among thousands of male students!

Slowly, the rules changed, and by 1907, Emmy had managed to become a student of math and even earn a doctorate—the highest qualification you can get. The next challenge was to get a job, at a time when most people still thought mathematicians had to be men.



Noether's theorem helps explain how air flows over a plane's wing.



Noether's theorem

When physicists are explaining strange things—such as how birds sit on power lines without getting electric shocks—they make use of Emmy's famous work, known as "Noether's theorem." Emmy used math, and this theorem, to describe symmetry in the world around us.

Emmy did her most famous research at the University of Göttingen, Germany.

